ENVX2001 Lab 07 - Regression modelling

ENVX2001 Applied Statistical Methods

Contents

Welcome to week 7!

This week's lab will give you some practice fitting and interpreting Simple Linear Regression (SLR) and Multiple Linear Regression (MLR) models.

This will prepare you for next week's content where we will seek to improve our models, and then in week 9 we will use improved models for prediction.

Learning outcomes

This module links to the following unit learning outcomes:

- LO3. demonstrate proficiency in the use or the statistical programming language R to apply an ANOVA and fit regression models to experimental data
- LO5. interpret the output and understand conceptually how its derived of a regression, ANOVA and multivariate analysis that have been calculated by R

- LO6. write statistical and modelling results as part of a scientific report

Specific goals

By the end of this lab, students should be able to:

- ☐ Fit a simple linear regression model
- ☐ Assess model assumptions and apply transformations when needed
- ☐ Interpret model parameters and fit statistics
- ☐ Fit a multiple linear regression model (**different structure**)
- ☐ Assess model assumptions, recalling key differences between MLR and SLR (**collinearity**)
- ☐ Determine the most suitable fit statistics for assessing the fit of a MLR model(**r-sq vs adj**)
- ☐ Write a conclusion on your findings

> Tip
>
> Please work on these exercises by creating your own Quarto file. You are welcome to write your responses up in the style of a report.

Preparation

- ☐ Install or update the `performance` package

CODE

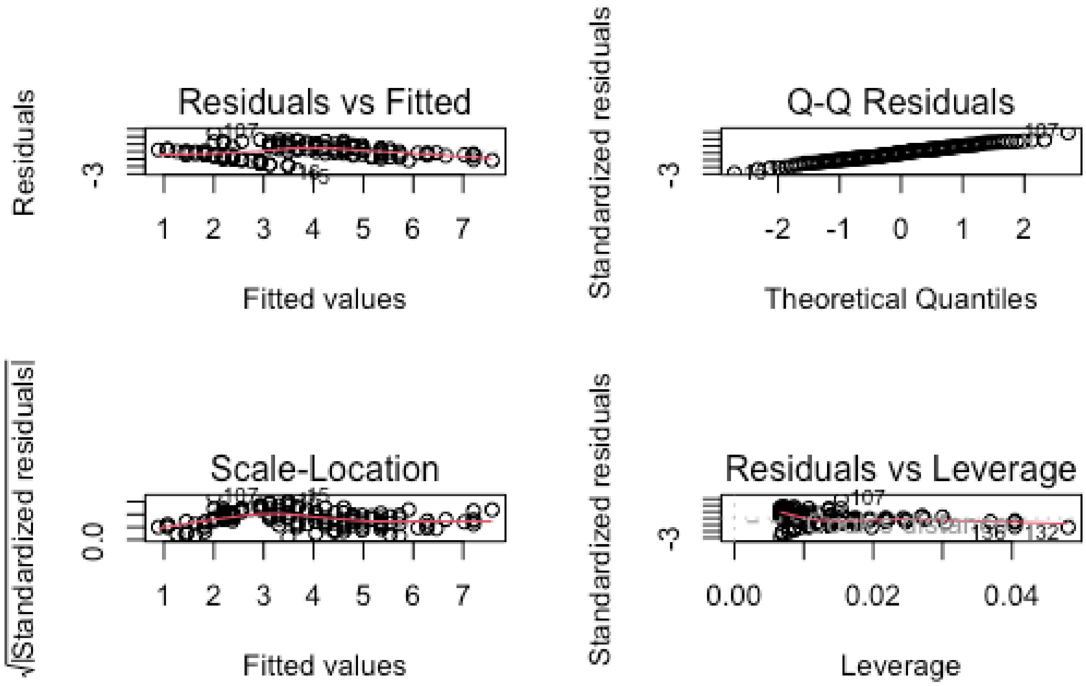
```r
#install.packages("performance")
library(performance)
```

This package is really good for checking your models. For this lab, we will focus on the `check_model()` function, which gives us nice pretty diagnostic plots for models:

`plot()`

CODE

```r
par(mfrow=c(2,2))
plot(iris_lm)
```



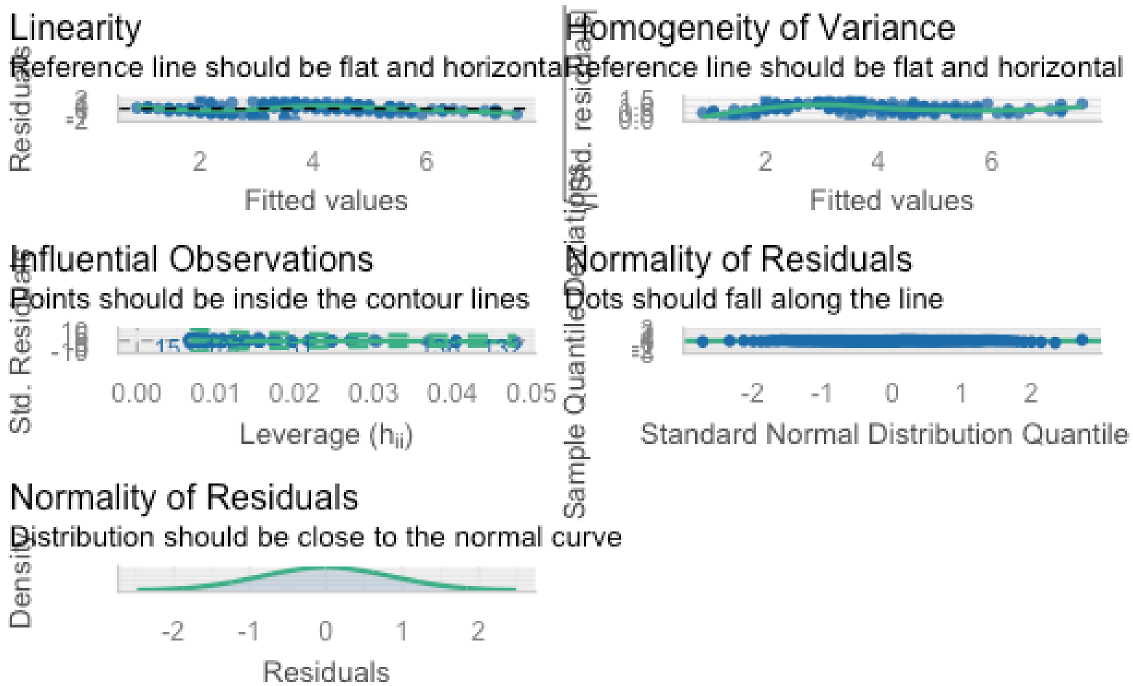
CODE

```
par(mfrow=c(1,1))
```

`check_model()` from `performance`

CODE

```
library(performance)
check_model(iris_lm)
```



Downloads

File	Used in	Download
`Loyn_SLR.xlsx`	Exercise 1	Download
`california_streamflow.xlsx`	Exercise 2	Download

Simple linear regression model fitting and interpretation (~ 30 minutes)

Simple linear regression is a statistical method that allows us to investigate the relationship between two continuous variables. It is a powerful tool for understanding how one variable (the predictor) influences another variable (the response). Often a simple linear regression is the most appropriate model, so it is worth understanding.

Key points to follow:

- State the model and hypotheses

- Conduct exploratory data analysis (EDA) to understand the data and relationships between variables. This includes correlation analysis and scatterplots.
- Fit the model
- Check the assumptions of the model and apply transformations if needed
- Interpret the model parameters and fit statistics

Good luck!

Analysis for bird abundance and area of forest patch

Figure 1: Kookaburra sitting on a branch. Photo by pen_ash on Unsplash

Habitat fragmentation resulting from agriculture and urbanisation has increased the rate of wildlife population decline in Australia. You are a consultant and have been asked to investigate the influence of forest fragment size upon bird abundance. You have been provided with some data containing data on bird abundance and patch area from 56 different forest patches in Southeastern Victoria, Australia, and you want to know if there is a relationship between these two variables.

The data is in the **Loyn_SLR.xlsx** file.

Variables are as follows:

- ABUND: Bird abundance within each forest patch (number of birds)
- AREA: Area of each forest patch (ha)

Original data source: Loyn, R. H. (1987). Effects of patch area and habitat on bird abundances, species numbers, and tree health in fragmented Victorian forests. In *Nature Conservation: the role of remnants of native vegetation* eds. D.S. Saunders, G.W. Arnold, A.A. Burbridge, &A.J.M. Hopkins, pp. 65-77. Surrey Beatty and Sons.

Exercise 1

(i) Write out the statistical model for a simple linear regression and define it in terms of influence of forest fragment size upon bird abundance. You can write it out in words and/or in mathematical notation.

(ii) State the hypotheses you will be testing for the relationship between bird abundance and patch area.

Remember: Null hypothesis = no change, no slope, therefore no relationship.

(iii) Conduct some exploratory data analysis upon your two variables. Describe the spread and central tendency of each, supported by values (summary statistics) and plots.

(iv) Investigate the relationship between bird abundance and patch area. Describe the relationship, supported by values (correlation coefficient) and plots.

(v) Fit a simple linear regression model to the data, with bird abundance as the response variable and patch area as the predictor variable.

(vi) Check the assumptions of your model. Do you need to transform any variables?

Hint 1: you can use the the `check_model()` function from the `performance` package to check the assumptions of your model.

Hint 2: If you do choose to transform a variable you will need to refit the model and check the assumptions again.

(vii) Interpret the model parameters and fit statistics. Recall the key information we draw from the output; what do these statistics tell you about the relationship between bird abundance and patch area?

(viii) Write a concluding statement on your findings. Hint: Consider the initial aim and hypothesis of the study and write it in a way that is easily understood by your client.

Multiple linear regression (~ 30 minutes)

Multiple linear regression has a slightly different approach compared to simple linear regression. We are still fitting a linear model and you will notice the process is similar, but we are incorporating more than one predictor variable into the model, which brings some additional steps.

- More hypotheses: rather than a single null and alternate hypothesis, you will also nee to consider the hypotheses for each predictor variable, as well as the overall model.
- More EDA: Rather than exploring only two variables, you will need to explore the additional variables. You will also need to explore the relationships between the predictor variables (signs of collinearity).
- More assumptions: in addition to the assumptions of linearity, independence, normality of residuals, and equal variance, we also need to consider the assumption of no collinearity between predictor variables as this can trick us into thinking our model is better than it actually is (we will explore this more in next week's lecture).
- More interpretation: Rather than only interpreting one regression coefficient, you will now have to assess a number of *partial regression coefficients* and their significance. You will also need to consider the overall model fit, and whether the model is a good fit for the data.
- You will need to use the adjusted r^2 value rather than the r^2 value to assess the fit of the model, as r^2 will always increase with the addition of more predictor variables, even if those variables don't actually improve the model fit. We will explain this more next week.

Analysis for runoff and precipitation in Bishop, California

Figure 2: Little Lakes Valley in the Eastern Sierra Nevada Mountains outside of Bishop, California. Source: Adobe Stock

You are a hydrologist tasked with analysing legacy data to investigate the relationship between precipitation at three sites (Lake Sabrina, Pine Creek, and Rock Creek) and stream runoff volume at a site near Bishop, California. The dataset contains 43 years of annual precipitation measurements (in mm) taken at (originally) 6 sites in the Owens Valley in California. You will focus on only 3 variables.

The data is in the **california_streamflow.xlsx** file.

Variables are as follows:

- `lake_sabrina`: Precipitation recorded at Lake Sabrina (mm)
- `pine_creek`: Precipitation recorded at Big Pine Creek (mm),
- `rock_creek`: Precipitation recorded at Rock Creek (mm)
- `runoff_volume`: Stream runoff volume measured at a site near Bishop, California (ML/year). This will be the response variable.

Note the variables have already been log-transformed to increase normality of the residuals in the regressions.

Data source: Hollett, K.J., Danskin, F.M., McCaffrey, W.F., and Walti, S.L., 1991. Geology and water resources of Owens Valley, California. U.S. Geological Survey Water-Supply Paper 2370-H.

> Tip
>
> This week we will focus on getting the model running and practice interpretation of the output. Next week we will focus on working out the most suitable model.

Exercise 2

(i) Write out the statistical model for a multiple linear regression and define it in terms of the data (precipitation at each site vs runoff volume at Bishop).

You can write it out in words and/or in mathematical notation.

(ii) State the hypotheses you will be testing for the relationship between each site and runoff volume.

(iii) Read in the data and conduct some exploratory data analysis upon your variables. Describe the spread and central tendency of each, supported by values (summary statistics) and plots.

(iv) Investigate the relationship between each of your variables. Describe the relationship, supported by values (correlation coefficient) and plots.

Hint: we are not only looking at relationship between response and predictors here, but also the relationships between predictor variables (signs of collinearity).

(v) Fit a multiple linear regression model to the data, with `runoff_volume` as the response variable and `lake_sabrina`, `pine_creek` and `rock_creek` as the predictor variables.

(vi) Check the assumptions of your model. Do you need to transform any variables?

> Hint 1: you can use the the `check_model()` function from the `performance` package to check the assumptions of your model.
>
> Hint 2: If you do choose to transform a variable you will need to refit the model and check the assumptions again.
>
> *(vii)* Interpret the model parameters and fit statistics. Recall the key information we draw from the output; what do these statistics tell you about the relationship between precipitation at each site and runoff volume at Bishop, California?
>
> *(viii)* Write a concluding statement on your findings. Hint: Consider the initial aim and hypothesis of the study and write it in a way that is easily understood by your client.

Closing thoughts

You have done a great job fitting linear regression models and interpreting the output.

As you may have noticed in this lab, models are rarely perfect, so next week we will focus on working out the most suitable model for our data based on what we have.

Keep up the great work and see you in the week 8 lecture!

If you have any questions, please approach your tutors or you are welcome to post to the Ed Discussion board.

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